FOURIER OPTICS

Advanced Labs II

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1 OBJECTIVES

The goal of this experiment is to understand Fourier optics experimentally. We aim to:

- Observe the Fourier plane of various objects
- Link spatial periodicity to frequency peaks
- Perform spatial filtering (low-pass and high-pass)
- Demonstrate image processing operations such as descreening and frequency selection

2 THEORY (MINIMAL)

2.1 Lens as Fourier transformer

In the paraxial approximation, the field at the focal plane of a lens is proportional to the 2D Fourier transform of the incident field.

Low spatial frequencies appear near the center, high frequencies further away.

Figure 1: Optical setup

2.2 Periodic object

For a structure of period Δ:

$$\delta = \frac{\lambda f_2}{\Delta}$$

If imaged on the camera with magnification M:

$$\delta' = M\delta = \frac{M\lambda f_2}{\Delta}$$

2.3 Role of aperture

The iris produces a convolution with the Airy disk:

$$\Phi \approx \frac{\lambda f_3}{d}$$

Reducing the aperture increases blur in Fourier space.

3 OBSERVATIONS IN THE FOURIER PLANE

3.1 Parameters

- Laser wavelength: $\lambda = 532\,\mathrm{nm}$
- Lens L_1: $20\,\mathrm{cm}$
- Lens L_2: $30\,\mathrm{cm}$

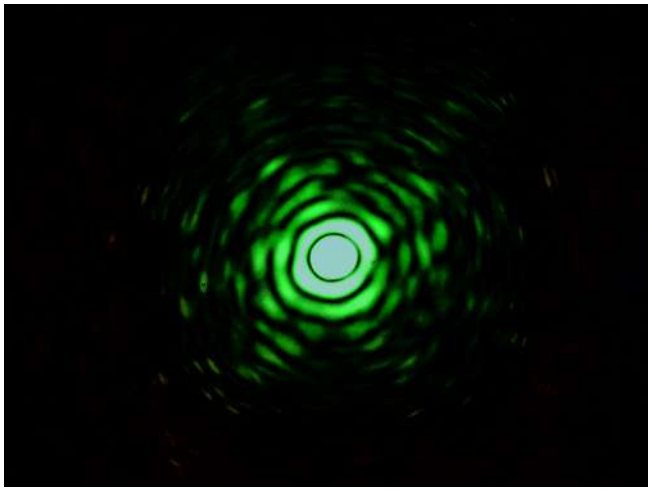

Figure 2: Airy Disk Obtained on Screen

- Lens L_3: 10 cm
- CMOS camera: pixel size 1.4 µm

3.2 Setup

Using the screen, we estimated the size of the non-augmented laser beam to be 3 mm. We then measured the diameter of the collimated beam the be 36 mm.

The laser beam first passes through a microscope objective (effective focal length $f_{\text{obj}} = 16\,\text{mm}$), then is collimated by lens L_1 of focal length $f_1 = 20\,\text{cm}$.

The objective produces a diverging beam with angular aperture $\theta \simeq \frac{D_{\text{in}}}{f_{\text{obj}}}$.

Which is then collimated to a beam radius $R = f_1\theta$,

hence we have that, for an input diameter $D_{\text{in}} = 3\,\text{mm}$,

$$D_{\text{out}} \simeq \frac{f_1}{f_{\text{obj}}} D_{\text{in}} = 3 \times \frac{200}{16} = 37.5\text{mm}$$

The relative deviation is therefore

$$\frac{|D_{\text{th}} - D_{\text{exp}}|}{D_{\text{th}}} = \frac{1.5}{37.5} \approx 4\%.$$

This is relatively small, and can be explained by experimental error. Notably, our collimation might have been slightly incorrect, our optical axis was maybe misaligned, but more importantly dominant source of error was probably the estimation of the initial beam diameter D_{in} from the screen image. Therefore, the result mainly validates the order of magnitude of the beam expansion rather than providing a high-precision verification.

3.3 Qualitative Observations

Here we place our screen in the Fourier plan and the iris bewteen our two lenses L1 and L2. With the screen placed in the Fourier plane, we would the expect the observed pattern to be the Fraunhofer diffraction pattern of the iris, i.e. the Fourier transform of a circular pupil, since collimation gives an image coming from infinity. Varying the radius of the iris, we make the following observations:

- Greater radius reduces diffraction. The airy disk and fringe spacing are smaller, and this makes sense since we are moving away from the diffraction limit, and at one point we simply end up with a magnified beam, as the fringes merge with the central fringe.
- Smaller radius increases diffraction. We have a larger disk and fringes further apart.

We then moved the iris close to L1 and put in place the OCD151 disk between the iris and lens L2. The OCD151 disk contains two combs of steps 500 µm and 850 µm.

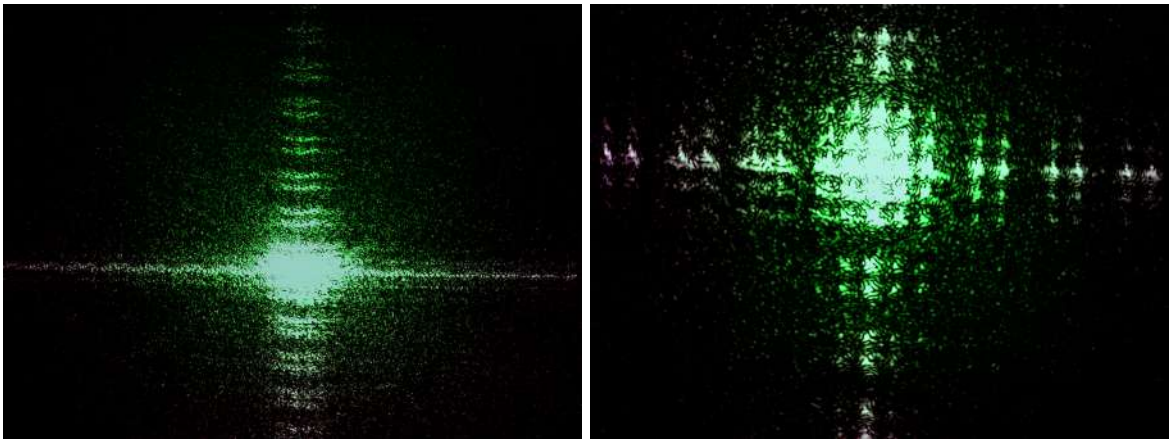

Figure 3: Fourier transforms of the OCD150 grids

For the OCD151 disk, we observed a pattern similr to a pattern we could obtain from diffraction grating. This can be interpreted using Fourier series: a periodic structure decomposes into discrete spatial frequencies, therefore its Fourier transform is a set of discrete peaks . The finite slit width produces a smooth sinc-type envelope, while the finite iris produces a circular Airy envelope. Hence the observed pattern corresponds to the convolution of the Fourier transform of the iris with that of the comb, which explains why each maximum appears broadened and modulated instead of being an infinitely thin line.

Rotation of the comb also rotated our pattren in the Fourier plane by the same angle. This makes sense because Fourier transform are covariant by rotation. We observed, as above, that pattern size was inversly proportional to the size of the diffracting object. The $500\,\mu$m comb therefore showed larger spacing than the $850\,\mu$m comb.

Replacing the disk with the OCD150 grid, we observed a two-dimensional lattice of bright spots. The symmetry along both horizontal and vertical directions makes sense because the grid can be seen as the superposition of two perpendicular comb gratings. Each direction generates its own discrete spatial frequencies, so then we have a 2D array of diffraction maxima modulated by a sinc envelope determined by the slit widths.

The pattern obtained using the smiley face was too small for us to observe anything.

3.4 Quantitative Measurements

For this part we analyzed the Fourier transform of several images images using a CMOS sensor.

To record the Fourier plane with the CMOS camera while keeping the possibility of inserting filters, we re-imaged the Fourier plane using an additional lens L_3 of focal length $f_3 = 10\,\text{cm}$. The instructions indicated this can be done by removing the disk and iris and putting the camera in the plane where the spot was the smallest and the brighest. This configuration introduces a magnification

$$M \approx \frac{D_1}{D_2},$$

.

With the iris alone (no object), the pattern in the Fourier plane is the diffraction pattern of a circular aperture so just an Airy disk. FRom that we know that the diameter to first zero is

$$\Phi \approx \frac{\lambda f_3}{d},$$

where λ is the laser wavelength and d the iris diameter. Closing the iris increased the Airy disk diameter (which makes sense from our formula) and ; opening the iris reduces the Airy disk and also improves resolution, but there we more aberrations. In our alignment we therefore tried to optimize the iris opening to obtain the sharpest patterns without overexposure and kept that opening for the rest of the measurements.

Next we wanted to check that the spacing between spots in Fourier space is inversely proportional to the real-space period, so a real-space period Δ leads in the Fourier plane (at the back focal plane of L_2) to a spot spacing

$$\delta = \frac{\lambda f_2}{\Delta}.$$

Since we re-image the Fourier plane onto the camera, the measured spacing on the sensor is

$$\delta' = M\delta = M\frac{\lambda f_2}{\Delta}.$$

Therefore, plotting or comparing δ' for two grids should confirm $\delta' \propto 1/\Delta$, and the magnification can be obtained from

$$M = \frac{\delta'\,\Delta}{\lambda f_2}.$$

The two grids provided by the guide have periods $\Delta_1 = 0.75\,\text{mm}$ and $\Delta_2 = 1.5\,\text{mm}$. Using ImageJ and the camera pixel size (ELP camera: $1.4\,\mu\text{m/pixel}$), we measured δ' over several consecutive periods to reduce uncertainty, then computed M from the relation above.For the grid of period $\Delta = 0.75\,\text{mm}$ we measure a spacing on the camera $\delta' = 199\,\mu\text{m}$. Using

$$M = \frac{\delta'\Delta}{\lambda f_2}, \qquad \lambda = 532\,\text{nm}, \quad f_2 = 30\,\text{cm},$$

we obtain

$$M = \frac{(1.99\times10^{-4})(7.5\times10^{-4})}{(5.32\times10^{-7})(0.30)} \simeq 0.94.$$

Moreover, comparing with the second grid ($\Delta = 1.5\,\text{mm}$, $\delta' = 97.2\,\mu\text{m}$), the product $\delta'\Delta \approx 1.49\times10^{-7}\,\text{m}^2$ is nearly identical in both cases, confirming the inverse proportionality $\delta' \propto 1/\Delta$ expected from Fourier optics.

For the grid of period $\Delta = 1.5\,\text{mm}$ we measure $\delta' = 97.2\,\mu\text{m}$. Applying the same relation gives

$$M = \frac{(9.72\times10^{-5})(1.5\times10^{-3})}{(5.32\times10^{-7})(0.30)} \simeq 0.91.$$

Again the value of $\delta'\Delta$ matches that of the smaller-period grid within a few percent, validating the theoretical scaling law. Averaging the two measurements yields an experimental magnification $M \approx 0.92$. The remaining discrepancy with the ideal value $M = 1$ is mainly attributed to uncertainties in locating the Fourier plane and in measuring peak positions on the camera. The pixel-to-distance calibration, limited resolution of the maxima, and slight misalignment of the optical elements all contributed to systematic errors but the Fourier scaling law was still validated experimentally. We noted that some spots of the Fourier transform of the grid fell in the zero of the Airy pattern of the iris, but we still took them into account for our computations.

Finally, we inserted the OCD140 disk directly in the Fourier plane between L_2 and L_3. When the OCD140 disk is correctly positioned, the camera image of the object shoiuld be sharp; we translated the disk along the rail until the slit edges were as sharp as we could do.

To measure the size of one slit, we used estimated magnification M of our "re-imaging" system A width measured on the sensor w_{cam} corresponds to a physical width in the Fourier plane

$$w_{\text{FP}} = \frac{w_{\text{cam}}}{M}.$$

Using ImageJ on the recorded image of the slit, we obtained a slit width on the order of

$$w_{\mathrm{FP}} \approx 3.0 \times 10^{-4}\,\mathrm{m} \;\; (\text{about } 300\,\mu\mathrm{m}),$$

with the dominant uncertainty coming from edge localization (blur/aberrations) and from the magnification calibration.

4 SPATIAL FILTERING

4.1 Descreening

In this part, we aim to remove the periodic stripes from the striped smiley (OCD150 disk) using Fourier filtering. The object is placed in the beam path and its Fourier transform is formed in the focal plane of lens L_2. A slit (OCD140 disk) is then positioned in this Fourier plane to suppress the spatial-frequency components associated with the stripes. The filtered image is finally observed after inverse Fourier transformation.

To switch between the Fourier plane and the image plane, the camera is translated along the optical axis. In our setup, lens L_3 was positioned at 131 cm. The Fourier plane was observed with the camera at 151.6 cm, while the sharp image was obtained for a camera position of 143.5 cm. These reference positions allowed us to move quickly between spectral observation and filtered image acquisition.

1. Reference images (image plane). The camera was first placed in the image plane (143.5 cm) to obtain sharp images of the four masks of the OCD150 disk (no filtering).

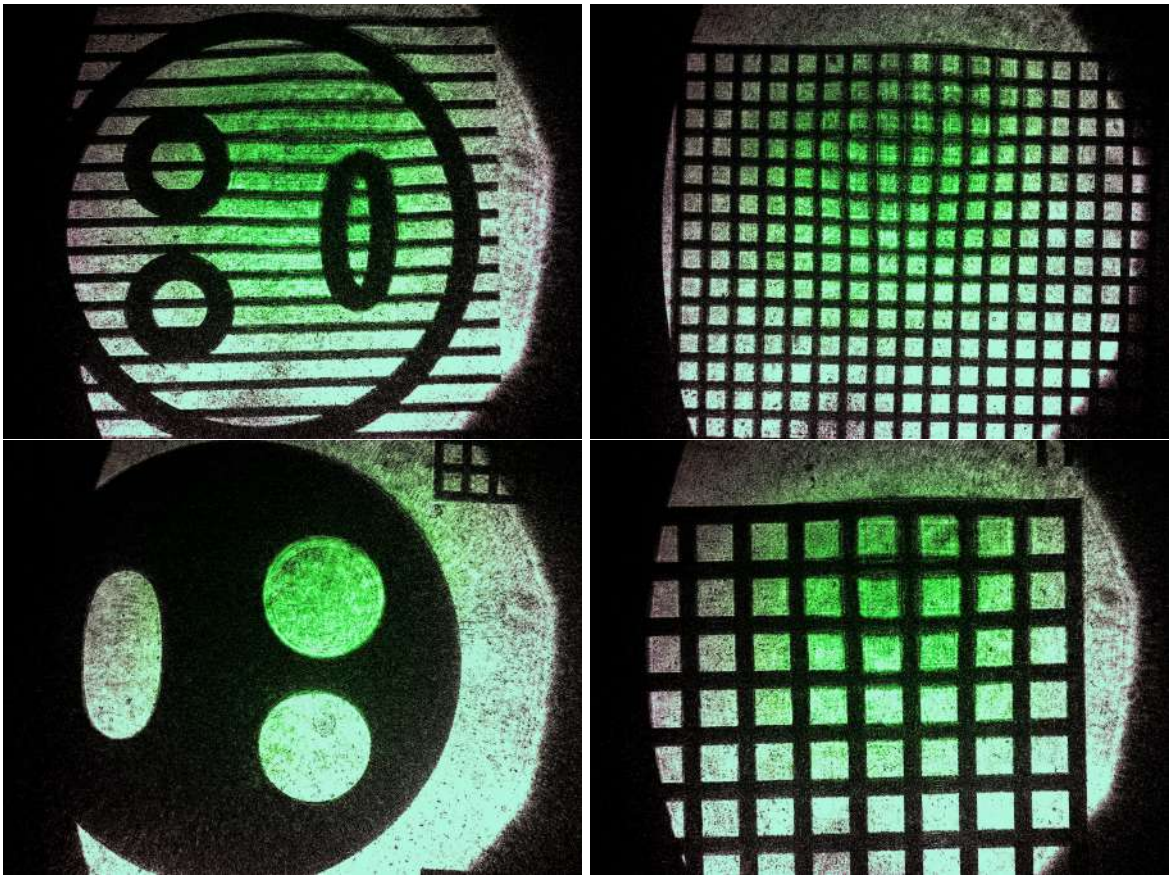

Figure 4: Reference images without filter.

The striped smiley clearly shows a periodic vertical pattern.

2. Filtering in the Fourier plane. The camera was moved to the Fourier plane (151.6 cm). The OCD140 disk was inserted and its thinnest slit was aligned to block the dots corresponding to the stripe frequency.

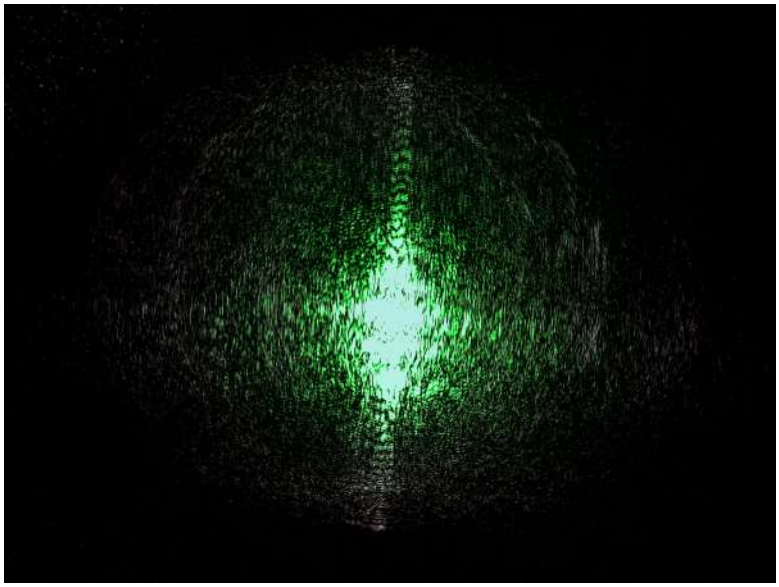

Figure 5: Fourier transform with slit blocking stripe frequencies.

3. Filtered image – thinnest slit. The camera was moved back to the image plane.

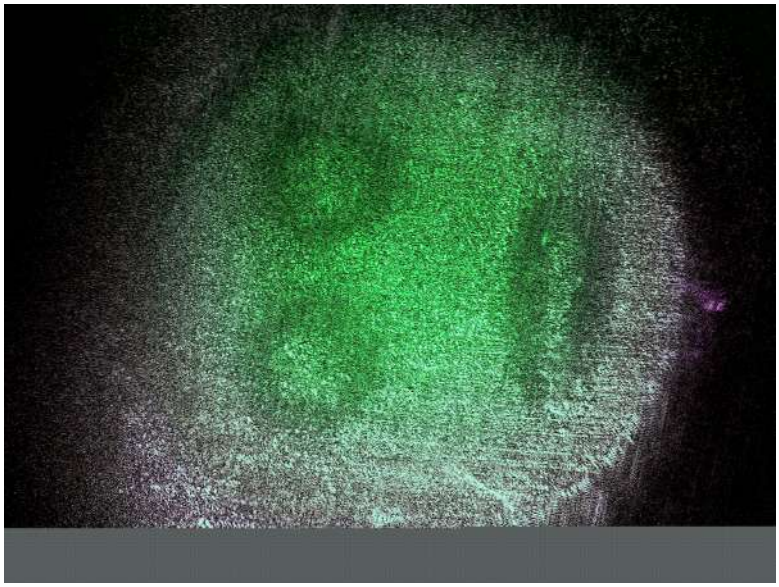

Figure 6: Filtered image (thinnest slit).

The stripes are strongly reduced and the smiley remains clearly visible.

4. Filtered image – thicker slit. The experiment was repeated with a thicker slit.

Figure 7: Filtered image (thicker slit).

With the thicker slit, residual stripes remain visible. The thinnest slit removes the stripe frequency more selectively. A thicker slit blocks a wider region of the Fourier plane, which reduces selectivity and makes the suppression less effective. The thinnest slit therefore gives better descreening.

4.2 Low-pass filter

In this part, a low-pass filter was implemented by placing a circular aperture (OCD130 disk) in the Fourier plane in order to keep only low spatial frequencies. The effect of the filter was first analysed on the laser beam and then on an image.

1. Identification of the Fourier center. The OCD150 disk was removed and the iris opened so that the incident beam corresponded to the laser beam only. Hole #5 of the OCD130 disk was aligned with this spot.

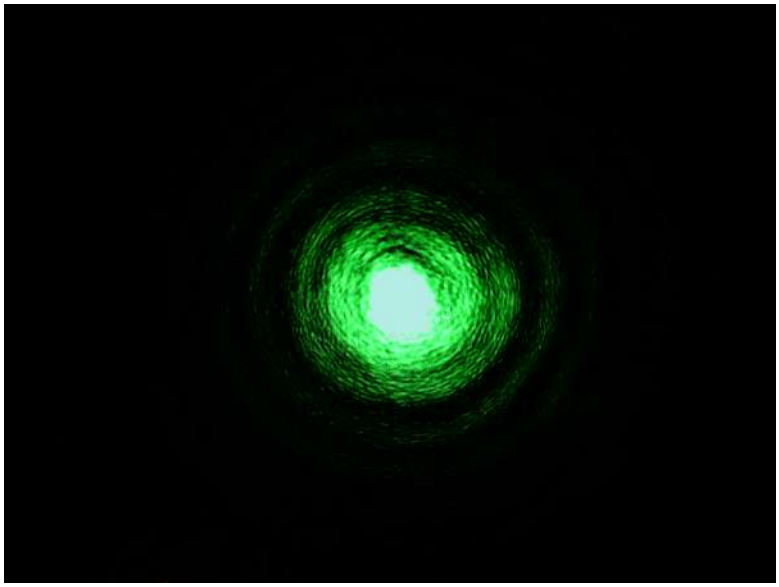

Figure 8: Hole #5 aligned with the center of the Fourier transform.

The diameter of hole #5 was measured to be approximately

$$d \simeq 300\,\mu\text{m}.$$

Since information close to the Fourier center corresponds to low spatial frequencies, this aperture acts as a low-pass filter.

2. Effect on the laser beam. Due to time constraints, we could not complete the imaging part of this laser beam. Here is our expected results however: When imaging the laser beam after inserting the hole, the transmitted beam is expected to become smoother and wider (like in the image shown in the lab subject). High spatial-frequency components associated with little intensity variations are erased, resulting in a homogeneous intensity profile. Increasing the hole diameter allows more frequencies to pass and should thus restores the original beam structure.

3. Filtering of the smiley image. The smiley was then placed back in the beam while keeping hole #5 in the Fourier plane.

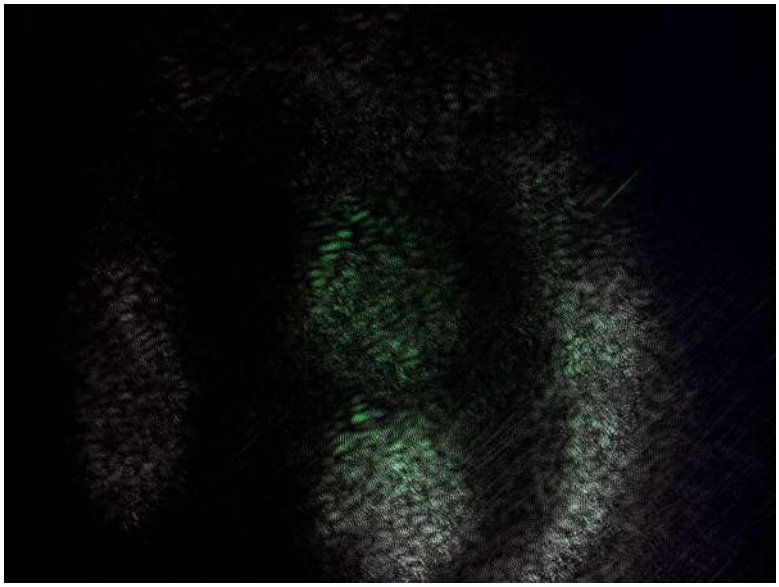

Figure 9: Smiley image after low-pass filtering.

The filtered image appears blurred. This is expected since high spatial frequencies, which are represented by sharp edges and fine details, are removed by the low-pass filter. Only big intensity variations are preserved, leading to loss of edges and reduced sharpness.

Overall, the low-pass filter smooths the image and erases details.

4.3 High-pass filter

In this part, a high-pass filter was implemented by blocking the central region of the Fourier transform in order to suppress low spatial frequencies and enhance image contours.

Filtering in the Fourier plane. The striped smiley (OCD150 disk) was placed in the laser beam. A dark disk from the OCD140 set was positioned in the Fourier plane to block the central spot corresponding to low spatial frequencies.

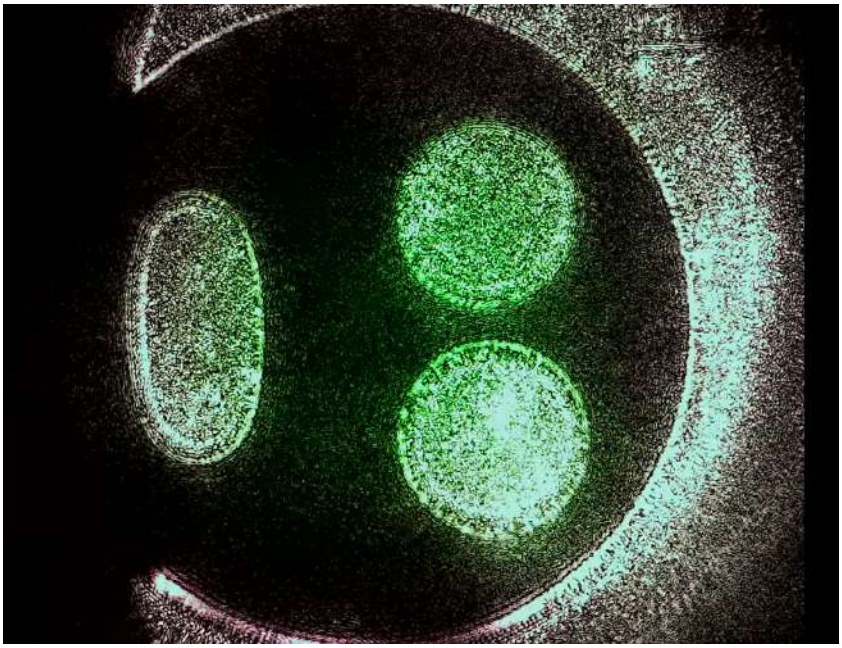

Figure 10: Smiley image after high-pass filtering.

After filtering, the contours and sharp intensity variations remain clearly visible. The smiley is mainly reconstructed through its edges.

This behavior is expected since low spatial frequencies contain intensity information, and high spatial frequencies correspond to rapid spatial variations such as edges. Blocking the Fourier center therefore enhances contours and suppresses smooth regions.

Increasing the diameter of the blocking disk strengthens this effect: more low frequencies are removed, leading to stronger edge enhancement but reduced overall brightness.

4.4 Frequency selection

For this section, due to time constraints, we could not perform any of the steps asked during the lab. We will however focus on the theory behind frequency doubling. A periodic comb of period Δ produces diffraction maxima satisfying

$$\Delta \sin\theta = m\lambda,$$

where m is the diffraction order. Each order corresponds to one spatial frequency component of the object.

If all diffraction orders contribute, the reconstructed image keeps the original period Δ.

When a Fourier filter selects only two symmetric orders $\pm m$, the field in the image plane can be written as

$$E(x) \propto e^{i2\pi fx} + e^{-i2\pi fx}, \qquad f = \frac{1}{\Delta}.$$

The measured intensity is then

$$I(x) \propto |E(x)|^2 \propto \cos^2(2\pi fx) = \tfrac{1}{2}\left(1 + \cos(4\pi fx)\right).$$

The dominant spatial frequency becomes $2f$, hence the new period is

$$\Delta_{\text{new}} = \frac{\Delta}{2}.$$

This corresponds to frequency doubling.

Numerical illustration

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(0,1,2000)
f = 5

I_before = 0.5*(1+np.cos(2*np.pi*f*x))
I_after  = 0.5*(1+np.cos(4*np.pi*f*x))

plt.plot(x,I_before,label="Original")
plt.plot(x,I_after,label="Doubled frequency")
plt.legend()
plt.xlabel("x")
plt.ylabel("Intensity")
plt.grid()
plt.show()
```

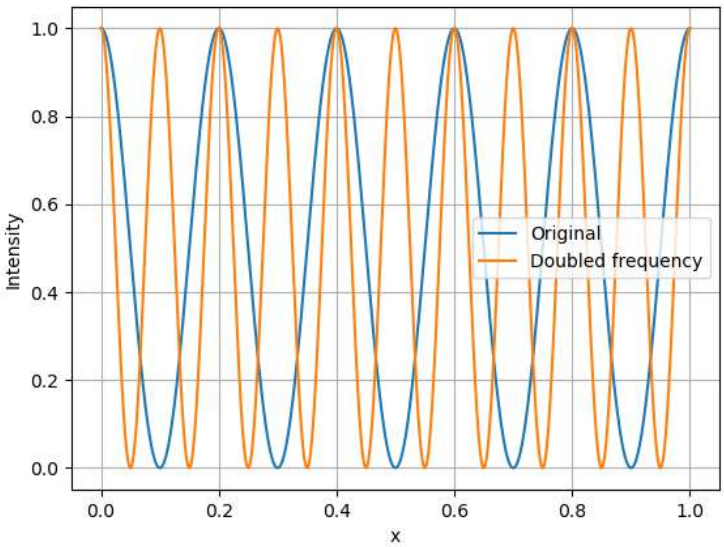

Figure 11: Intensity profile.

We can observe that the fringe spacing in the original image is double the fringe spacing of the filtered image, which corresponds to a frequency doubling.

CONCLUSION

In this lab, we verified experimentally the main principles of Fourier optics. We observed that a lens performs the two-dimensional Fourier transform of an object in its focal plane and that spatial frequencies are directly accessible in this Fourier plane. Qualitative and quantitative measurements confirmed the inverse relation between object period and spacing of diffraction peaks, consistent with $\delta' = M\lambda f_2/\Delta$. By inserting masks in the

Fourier plane, we selectively removed spatial frequencies and demonstrated low-pass filtering (image smoothing), high-pass filtering (edge enhancement), descreening of periodic patterns, and frequency doubling through order selection.